

Targeting methionine auxotrophy in cancer: discovery & exploration.

INTRODUCTION: Amino acid auxotrophy or the metabolic defect which renders cancer incapable of surviving under amino acid depleted conditions is being exploited and explored as a therapeutic against cancer. Early clinical data on asparagine- and arginine-depleting drugs have demonstrated low toxicity and efficacy in melanoma, hepatocellular carcinoma and acute lymphoblastic leukemia. Methionine auxotrophy is a novel niche currently under exploration for targeting certain cancers. **AREAS COVERED:** In this review we explore the discovery of methionine auxotrophy followed by in vitro, in vivo and patient data on targeting cancer with methionine depletion. We end with a small discussion on bioengineering, pegylation and red blood cell encapsulation as mechanisms for decreasing immunogenicity of methionine-depleting drugs. We hope to provide a platform for future pharmacology, toxicology and cytotoxicity studies with methionine depletion therapy and drugs. **EXPERT OPINION:** Although methionine auxotrophy seems as a viable target, extensive research addressing normal versus cancer cell toxicity needs to be conducted. Further research also needs to be conducted into the molecular mechanism associated with methionine depletion therapy. Finally, novel methods need to be developed to decrease the immunogenicity of methionine-depleting drugs, a current issue with protein therapeutics.